

At the Critical Line

Continued fractions and sharp Riesz summation of resonant transseries

Research article prepared for Vladimir Reshetnikov
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Abstract

The residue expansion of a meromorphic Borel transform can diverge even when its Stokes jump is holomorphic and has a normally convergent finite-block representation. We resolve the critical-line question explicitly left open in the *ProveIt* resonance-block report for two sine lattices with numerator t^D , $D \geq 2$. If p_k/q_k are the convergents of an irrational period ratio, $b = \limsup \log q_{k+1}/q_k \in (0, \infty)$, and $A_k = q_k^D q_{k+1} e^{-bq_k}$, then the increasing-action residue series converges on the entire line $\operatorname{Re} z = b$ exactly when $A_k \rightarrow 0$, and converges absolutely exactly when $\sum A_k < \infty$. The two separately indexed lattice series have a different exact criterion: convergence of $\sum (-1)^{p_k+q_k+k} A_k e^{-iq_k \operatorname{Im} z}$. The proof removes an absolutely summable remainder and isolates the primitive convergent pairs. Continued-fraction constructions realize absolute convergence, conditional convergence, bounded divergent spikes, and unbounded spikes at every prescribed positive critical abscissa. A derivative hierarchy shows precisely when an analytic sum permits termwise differentiation of its raw representation.

For d sine lattices we also prove that Riesz weighting of integer order $m \geq d - 1$ recovers the block sum throughout $\operatorname{Re} z > 0$, without any Diophantine hypothesis. It has an exact finite derivative bias and an exponentially small remainder, uniform across arbitrarily close poles. The threshold $d - 1$ is sharp for gap-independent uniform control, while continuity at a collision exactly on the cutoff requires one additional order. Richardson extrapolation removes the bias, and a local inverse stability theorem transports the exponential accuracy through nonlinear inversion. All new assertions are proved conventionally; 192 exact finite checks and 49 high-precision diagnostics accompany the article. No Lean verification or independently established publication priority is claimed.

Inspected repository snapshot:

3d5973524506411392a911470b5ddc35521568ea.

The source, PDF, verification code, recorded data, and build instructions are distributed together. No repository files were modified.

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1 The open question and the results

1.1 The precise repository frontier

The *ProveIt* report *Resonance-Block Summation of Transseries* [1] studies

$$\widehat{F}_\omega(t) = \frac{t^d}{\prod_{j=1}^d \sin(\pi \omega_j t)}, \quad \omega_j > 0. \quad (1.1)$$

It proves a separation-independent block representation of the Stokes jump, identifies the open half-plane of absolute convergence for two irrationally related lattices, and transports the block sum through analytic inversion. These are antecedents, not results claimed here for the first time.

Its further-research Question 1 asks for a continued-fraction criterion on the boundary of that half-plane, distinguishing separate lattice summation from increasing-action summation and constructing the possible convergence types. The main result below answers that question. The comparison is to the pinned source, especially its labels `thm:block`, `thm:arithmetic`, and Section `sec:questions`, Question 1. The repository inventory [2] was also checked; this is not an exhaustive priority audit of every document in the repository or in the literature.

There is a useful distinction behind the problem. A complete close pair has a bounded sum, but either constituent can be enormous. A cutoff between the two poles reveals one constituent. Whether those unfinished pairs vanish is a different question from whether all their absolute values are summable. Continued fractions identify exactly which pairs matter at the critical line.

1.2 An overview in the original degree-two model

For $(\omega_1, \omega_2) = (1, \alpha)$ with $\alpha > 0$ irrational, put

$$b = \beta(\alpha) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log \frac{1}{\text{dist}(n\alpha, \mathbb{Z})}. \quad (1.2)$$

The predecessor proves absolute convergence for $\text{Re } z > b$ and failure of the term test for $\text{Re } z < b$. For $0 < b < \infty$, let p_k/q_k be the continued-fraction convergents and set

$$A_k = q_k^2 q_{k+1} e^{-b q_k}, \quad \eta_k = (-1)^{p_k + q_k + k}. \quad (1.3)$$

Our increasing-action classification on $\text{Re } z = b$ is

Condition on A_k	Raw residue series, ordered by action
$\sum_k A_k < \infty$	Absolutely convergent.
$A_k \rightarrow 0$ and $\sum_k A_k = \infty$	Conditionally convergent.
$A_k \not\rightarrow 0$	Divergent; its terms do not tend to zero.

The answer is the same at every imaginary height. By contrast, the two separate lattice series converge at $b + iy$ precisely when $\sum_k \eta_k A_k e^{-iy q_k}$ converges. In particular, an increasing-action sum can converge although its separate lattice sums are respectively $+\infty$ and $-\infty$ at the real boundary point.

The second part of the article supplies a different repair when ordinary summation fails. Write λ for an action and c_λ for its Borel residue. For two lattices,

$$\mathcal{R}_T^{(1)}(z) = \sum_{\lambda < T} \left(1 - \frac{\lambda}{T}\right) c_\lambda e^{-\lambda z} = \mathcal{S}(z) + \frac{\mathcal{S}'(z)}{T} + O_K(T e^{-\sigma T}), \quad (1.4)$$

where $K \subseteq \{\text{Re } z > 0\}$ and $\sigma = \min_K \text{Re } z$. Thus $2\mathcal{R}_{2T}^{(1)} - \mathcal{R}_T^{(1)}$ approximates the normalized Stokes jump $\mathcal{S}$ with exponentially small error. This works even when $\beta(\alpha) = \infty$. It does not require finding the close pairs before defining the weighted finite sum.

1.3 Established ingredients and the novelty boundary

Continued-fraction approximation and the exponential approximation exponent are classical. In particular, Sondow's irrationality base has logarithm $\limsup \log q_{k+1}/q_k$ [3]; our b is that *logarithm*, not the base itself. Divided differences and their Genocchi–Hermite representation are classical [4]. Riesz means of general Dirichlet series have a substantial theory; the work of Defant and Schoolmann gives a modern primary reference [5]. Richardson extrapolation is ordinary polynomial interpolation at zero in the variable $1/T$; we prove the needed finite identities instead of attributing novelty to that procedure.

The additions relative to the inspected repository report are the exact critical-line reduction, its ordering-sensitive realization theory, the raw derivative hierarchy, and the sharp cutoff-regularization theorem with its finite bias and exponential acceleration. These are model-specific results, not a general solution to all summability questions for transseries. Their proofs stand independently of a claim to being the first occurrence of every refinement in the literature. The computational checks are supplementary evidence, not substitutes for the proofs.

2 The Borel model, actions, and block summation

2.1 Definitions and conventions

Fix integers $d \geq 2$ and $D \geq d$. We use the slight extension

$$\hat{H}_{\omega,D}(t) = \frac{t^D}{\prod_{j=1}^d \sin(\pi\omega_j t)}, \quad \mathbb{H} = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}. \quad (2.1)$$

The singularity at zero is removable. The original model is $D = d$; allowing D to vary makes differentiation transparent. To make the transseries connection explicit, expand the analytic germ at zero as $\hat{H}(t) = \sum_{n \geq 0} h_n t^n$. In the Borel convention used here it corresponds to the formal inverse-power series

$$\tilde{H}(z) = \sum_{n \geq 0} n! h_n z^{-n-1}. \quad (2.2)$$

Its lateral Laplace realizations differ by the exponential residue corrections studied below. A finite union of positive action lattices is locally finite and lies in the additive monoid generated by their positive spacings. Thus the obstruction is not an invalid formal action support: it is failure of an infinite raw numerical sum at a fixed argument. Finite-block or Riesz realization repairs the latter without changing the input actions.

Until the confluence discussion, assume that ω_i/ω_j is irrational for $i \neq j$. All positive poles are then simple. At $\lambda = n/\omega_j$ their residues are

$$c_{n/\omega_j} = \frac{(-1)^n (n/\omega_j)^D}{\pi\omega_j \prod_{\ell \neq j} \sin(\pi\omega_\ell n/\omega_j)}. \quad (2.3)$$

The action set Λ is the union of these d positive lattices. It is locally finite. Define the increasing-action cutoff by

$$P_T(z) = \sum_{\substack{\lambda \in \Lambda \\ \lambda < T}} c_\lambda e^{-\lambda z}. \quad (2.4)$$

Ordinary increasing-action convergence means that $P_T(z)$ has a limit as $T \rightarrow \infty$, with no restriction that T avoid a small gap in a particular way. The strict inequality fixes values at poles but does not affect the convergence question.

For two lattices we write the two separate series as

$$r_n(z) = \frac{(-1)^n n^D e^{-nz}}{\pi \sin(\pi \alpha n)}, \quad R(z) = \sum_{n \geq 1} r_n(z), \quad (2.5)$$

$$s_m(z) = \frac{(-1)^m (m/\alpha)^D e^{-(m/\alpha)z}}{\pi \alpha \sin(\pi m/\alpha)}, \quad S(z) = \sum_{m \geq 1} s_m(z). \quad (2.6)$$

The capital letters R, S in this display refer only to the separate series. The calligraphic $\mathcal{S}$ below always denotes the block sum.

2.2 Uniform blocks: a recalled mechanism with proof

Fix a compact positive period box

$$0 < m_0 \leq \omega_j \leq M_0, \quad h = \frac{1}{8(d+1)M_0}. \quad (2.7)$$

Join successive pole occurrences when their gap is less than h . The connected components are called blocks. When a repeated pole is allowed, its occurrences belong to one block and are counted with multiplicity.

Lemma 2.1 (Geometry and pole-cancelled factors). *Every block C has at most d occurrences and interval length at most $(d-1)h$. Consecutive blocks are separated by at least h . Their minima have a uniformly bounded count in every unit interval. Put $a_C = \min C$ and*

$$g_C(t, z) = e^{-zt} \hat{H}_{\omega, D}(t) \prod_{\lambda \in C} (t - \lambda). \quad (2.8)$$

There is a fixed tube around each real block interval on which g_C is holomorphic. For every compact $K \Subset \mathbb{H}$ and every fixed integer $J \geq 0$, constants independent of internal pole gaps satisfy

$$\sup_{\substack{z \in K \\ t \in [\min C, \max C]}} |\partial_t^j g_C(t, z)| \leq C_{K, J} (1 + a_C)^D e^{-\sigma a_C}, \quad 0 \leq j \leq J, \quad \sigma = \min_K \operatorname{Re} z. \quad (2.9)$$

Proof. If $d+1$ successive occurrences belonged to a block, their first d gaps would give a span less than $dh < 1/M_0$. Two occurrences would come from one lattice, contradicting its spacing $1/\omega_j \geq 1/M_0$. This also shows that a block contains at most one occurrence from each lattice. The separation statement is the definition. A unit interval contains at most $d(M_0 + 1)$ pole occurrences, rounded upward if necessary, which bounds its number of block minima.

Take a tube of radius, for example, $h/4$ around the block interval. An uncanceled positive zero is at distance at least h from the interval. The tube also stays away from zero and the negative poles. For a lattice represented by a node $\lambda = n/\omega_j$, its cancelled factor is, up to sign,

$$\frac{t - \lambda}{\sin(\pi \omega_j (t - \lambda))}.$$

It is uniformly bounded on a slightly larger tube: the argument remains strictly inside the distance to the next zero, and its removable value is $1/(\pi \omega_j)$. For a lattice not represented in the block, the sine factor is bounded away from zero on that tube. These assertions follow either from $|\sin(u + iv)|^2 = \sin^2 u + \sinh^2 v$ or from compactness after reduction modulo its period. The period box supplies uniform constants.

The numerator is bounded by a constant times $(1 + a_C)^D$. On the tube, $|e^{-zt}| \leq C_K e^{-\sigma a_C}$, since its width and the block length are fixed and z ranges over K . Cauchy's estimates on a nested tube give every fixed derivative bound in (2.9). No step divides by an internal gap. $\square$

For distinct real nodes $\lambda_1, \dots, \lambda_r$, write

$$[\lambda_1, \dots, \lambda_r]f = \sum_{i=1}^r \frac{f(\lambda_i)}{\prod_{j \neq i} (\lambda_i - \lambda_j)}. \quad (2.10)$$

The divided difference has its usual confluent meaning at repeated nodes. The Genocchi–Hermite formula, with simplex volume $1/(r-1)!$, gives

$$|[\lambda_1, \dots, \lambda_r]f| \leq \frac{\sup_I |f^{(r-1)}|}{(r-1)!}, \quad I = [\min \lambda_i, \max \lambda_i]. \quad (2.11)$$

This follows by iterating the fundamental theorem of calculus; see also [4, formula (52)]. The same estimate holds for distinct nodes when $f \in W^{r-1, \infty}(I)$, with the essential supremum on the right. Indeed, extend f across the endpoints using its derivative traces, mollify, apply the smooth estimate, and pass to the uniform limit at the finitely many nodes. For $r = 1$ it is simply a bound on a value. We will need this Sobolev version at a nonsmooth cutoff, but will not use it to assign an undefined derivative to a fully confluent node.

Proposition 2.2 (Block sum and the Stokes jump). *For a block C , let*

$$B_C(z) = \sum_{\lambda \in C \text{ distinct}} \text{Res}_{t=\lambda} (e^{-zt} \widehat{H}(t)) = [\lambda_1, \dots, \lambda_r]g_C(\cdot, z). \quad (2.12)$$

Then $\mathcal{S}(z) = \sum_C B_C(z)$ converges absolutely and locally uniformly on $\mathbb{H}$, together with each fixed z -derivative. It is independent of the sufficiently small admissible block threshold. With lower and upper lateral Laplace integrals F_-, F_+ ,

$$F_-(z) - F_+(z) = 2\pi i \mathcal{S}(z). \quad (2.13)$$

All block estimates are uniform on the period box, including confluent blocks.

Proof. The residue identity follows from the contour formula for divided differences. Equations (2.9) and (2.11) majorize $|B_C(z)|$ by $C_K(1 + a_C)^D e^{-\sigma a_C}$. The bound on the number of minima in a unit interval makes the sum finite. Multiplying the numerator by any fixed power of t proves the derivative assertion.

For completeness, choose small rays above and below the positive axis such that $\text{Re}(ze^{i\theta}) > 0$ along the intervening angles. The Laplace integrals converge at zero because $D \geq d$, and at infinity because the sine factors grow in either open half-plane. Close the rays at radii in the middle of the gaps between blocks. These radii stay a fixed distance from real poles. On the closing arcs, the sine factors are bounded away from zero near the real axis and grow away from it; the numerator is at most polynomial. The exponentially decaying Laplace factor makes the outer arc tend to zero. The residue theorem gives (2.13), with the lower ray traversed outward. Every admissible blocking has the same two lateral integrals, which proves independence. This is the predecessor's block argument with D in place of d [1]. $\square$

3 Continued fractions and exponentially weighted small divisors

3.1 The approximation facts, including their proofs

Write $\alpha = [a_0; a_1, a_2, \dots]$ and use

$$\begin{aligned} p_{-2} &= 0, & p_{-1} &= 1, & p_k &= a_k p_{k-1} + p_{k-2}, \\ q_{-2} &= 1, & q_{-1} &= 0, & q_k &= a_k q_{k-1} + q_{k-2}. \end{aligned}$$

The index k always retains this convention; it determines a sign in the main theorem. Finitely many initial exceptional indices, including a repeated initial denominator or a zero numerator, may be deleted from summability statements without renumbering the remaining indices. Set $\delta_k = \alpha q_k - p_k$.

Lemma 3.1 (Continued-fraction estimates). *For all sufficiently large k ,*

$$\operatorname{sgn} \delta_k = (-1)^k, \quad q_{k+1} < |\delta_k|^{-1} < q_{k+1} + q_k. \quad (3.1)$$

For $q_k \leq n < q_{k+1}$,

$$\|n\alpha\| \geq |\delta_k|, \quad \|u\| := \operatorname{dist}(u, \mathbb{Z}). \quad (3.2)$$

If integers p, n , $n > 0$, satisfy $|n\alpha - p| < 1/(2n)$, then the reduced fraction p/n is a convergent. Finally, q_k grows at least geometrically in k after grouping adjacent indices.

Proof. Let $\tau_{k+1} = a_{k+1} + \theta_k$ be the complete quotient, where $0 < \theta_k < 1$. The recurrence gives

$$\alpha = \frac{p_k \tau_{k+1} + p_{k-1}}{q_k \tau_{k+1} + q_{k-1}}, \quad \delta_k = \frac{(-1)^k}{q_k \tau_{k+1} + q_{k-1}},$$

since $p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1}$. This proves (3.1) and $\delta_{k-1} = -\tau_{k+1} \delta_k$.

To prove the best-approximation assertion, suppose $0 < n < q_{k+1}$ and $|n\alpha - p| < |\delta_k|$. The determinant identity writes $(p, n) = u(p_k, q_k) + v(p_{k-1}, q_{k-1})$ with $u, v \in \mathbb{Z}$. Thus $|u - v\tau_{k+1}| < 1$. If $v > 0$, this implies $u \geq va_{k+1}$ and $n \geq vq_{k+1}$, a contradiction. If $v < 0$, it implies $u \leq va_{k+1}$ and $n \leq vq_{k+1} < 0$. If $v = 0$, it forces $u = 0$. All possibilities are impossible. Taking the nearest p proves (3.2); eventually p_k itself is nearest to $q_k \alpha$.

For the Legendre assertion, first reduce $p/n = P/Q$. Its error is less than $1/(2Q^2)$. Choose k with $q_k \leq Q < q_{k+1}$. Unless $P/Q = p_k/q_k$, best approximation and the determinant bound give

$$1 \leq |Pq_k - Qp_k| \leq q_k |Q\alpha - P| + Q|\delta_k| \leq 2Q|Q\alpha - P| < 1,$$

a contradiction. Initial small-denominator cases follow from the same continued-fraction identities, or can be omitted in our tail arguments. Finally $q_{k+2} \geq q_{k+1} + q_k \geq 2q_k$ once denominators increase. $\square$

Corollary 3.2 (Exponential exponent and reciprocity). *For every positive irrational α ,*

$$\beta(\alpha) = \limsup_k \frac{\log q_{k+1}}{q_k}, \quad \alpha\beta(1/\alpha) = \beta(\alpha), \quad (3.3)$$

including the values zero and infinity.

Proof. Evaluating at $n = q_k$ and using (3.1) proves one inequality in the first formula. For $q_k \leq n < q_{k+1}$, best approximation bounds $n^{-1} \log(1/\|n\alpha\|)$ above by $q_k^{-1} \log(q_{k+1} + q_k)$. The difference between this and $q_k^{-1} \log q_{k+1}$ tends to zero. This proves equality. It agrees with the classical irrationality-base formula [3].

Apart from initial terms, the convergents of $1/\alpha$ are q_k/p_k . Since $p_k/q_k \rightarrow \alpha$ and $\log p_{k+1} = \log q_{k+1} + O(1)$, the second equality follows, with extended-real limits understood. $\square$

3.2 An exact positive-series and term-test reduction

Lemma 3.3 (Weighted denominator test). *Let $x > 0$ and $D \geq 0$ be fixed, and set $A_k(x) = q_k^D q_{k+1} e^{-xq_k}$. Then*

$$\sum_{n \geq 1} \frac{n^D e^{-xn}}{|\sin(\pi n \alpha)|} < \infty \iff \sum_k A_k(x) < \infty, \quad (3.4)$$

$$\frac{n^D e^{-xn}}{|\sin(\pi n \alpha)|} \longrightarrow 0 \iff A_k(x) \longrightarrow 0. \quad (3.5)$$

Proof. Use $2\|u\| \leq |\sin(\pi u)| \leq \pi\|u\|$. For the integer block $q_k \leq n < q_{k+1}$, Lemma 3.1 gives the upper bound

$$\begin{aligned} \sum_{n=q_k}^{q_{k+1}-1} \frac{n^D e^{-xn}}{|\sin(\pi n \alpha)|} &\leq \frac{q_{k+1} + q_k}{2} \sum_{n \geq q_k} n^D e^{-xn} \\ &\leq C_{x,D} q_k^D q_{k+1} e^{-xq_k}. \end{aligned}$$

Here $n = q_k + j$ and $(q_k + j)^D \leq q_k^D (1 + j)^D$ bound the last sum. The single convergent index $n = q_k$ is bounded below by $A_k(x)/\pi$. The denominator blocks partition a tail of the positive integers. These inequalities prove (3.4). They also bound every term in the k th block by $C_{x,D} A_k(x)$, proving both directions of (3.5). $\square$

This lemma already identifies the absolute convergence criterion on a critical line. Conditional convergence requires more information: the near-cancelling residues must be matched and their signs retained. That is the purpose of the next section.

4 Critical-line extraction and the complete ordering criterion

Assume throughout this section that $D \geq 2$, $\alpha > 0$ is irrational, and $0 < b = \beta(\alpha) < \infty$. Define

$$A_k = q_k^D q_{k+1} e^{-bq_k}, \quad \eta_k = (-1)^{p_k + q_k + k}, \quad \lambda'_k = p_k/\alpha = q_k - \delta_k/\alpha. \quad (4.1)$$

The two actions q_k and λ'_k will be called the primitive convergent pair.

4.1 The absolutely summable remainder

Lemma 4.1 (Nonprimitive residues are summable). *After deleting r_{q_k} from the first lattice and s_{p_k} from the second lattice, the remaining residue terms on $\operatorname{Re} z = b$ are absolutely summable. The sums of their absolute values are independent of $\operatorname{Im} z$.*

Proof. First consider an integer n in the first lattice. If $\|n\alpha\| \geq 1/(2n)$, then $|r_n(b + iy)| \leq n^{D+1} e^{-bn}/\pi$, and these bounds sum. Otherwise choose the nearest integer p and reduce p/n . By Lemma 3.1,

$$n = \ell q_k, \quad p = \ell p_k, \quad \|n\alpha\| = \ell |\delta_k|$$

for a unique reduced convergent and a positive integer ℓ . A nonprimitive term has $\ell \geq 2$. Its absolute value is at most

$$C q_k^D (q_{k+1} + q_k) \ell^{D-1} e^{-b\ell q_k}.$$

The sum over $\ell \geq 2$ is at most $C_{b,D} q_k^D (q_{k+1} + q_k) e^{-2bq_k}$. Because $\limsup \log q_{k+1}/q_k = b$, eventually $q_{k+1} \leq e^{3bq_k/2}$. Thus the last bound is summable in k , using geometric growth of q_k . Finally many early k cause no problem, since their sums over ℓ converge.

Apply the same proof to $1/\alpha$ with the Laplace parameter $b/\alpha = \beta(1/\alpha)$. Its primitive denominators are p_k , apart from initial terms. The constant factor α^{-D-1} in the second lattice does not affect summability. Absolute values remove y from both arguments. $\square$

Theorem 4.2 (Critical-line extraction modulo ℓ^1). *For every $Y < \infty$ there are functions u_k, v_k on $[-Y, Y]$ such that*

$$r_{q_k}(b + iy) = \frac{\eta_k}{\pi^2} A_k e^{-iyq_k} + u_k(y), \quad (4.2)$$

$$s_{p_k}(b + iy) = -\frac{\eta_k}{\pi^2} A_k e^{-iyq_k} + v_k(y), \quad (4.3)$$

$$\sum_k \sup_{|y| \leq Y} (|u_k(y)| + |v_k(y)|) < \infty. \quad (4.4)$$

Together with Lemma 4.1, these formulas account for all but finitely many raw residues.

Proof. Let $F_z(t) = t^D e^{-zt}$. The first residue has the form

$$r_{q_k}(z) = \frac{(-1)^{p_k+q_k} F_z(q_k)}{\pi \sin(\pi \delta_k)}.$$

For small real δ , $1/\sin(\pi\delta) = 1/(\pi\delta) + O(|\delta|)$. Moreover $|\delta_k|^{-1} = q_{k+1} + \theta_k q_k$, $0 < \theta_k < 1$. Using $\operatorname{sgn} \delta_k = (-1)^k$ gives (4.2), with

$$\sup_{|y| \leq Y} |u_k(y)| \leq C_Y q_k^{D+1} e^{-bq_k} \quad (4.5)$$

for large k .

For the companion residue, an exact sine identity gives

$$s_{p_k}(z) = -\frac{(-1)^{p_k+q_k} F_z(q_k - \delta_k/\alpha)}{\pi \alpha \sin(\pi \delta_k/\alpha)}.$$

Replace $1/(\pi \alpha \sin(\pi \delta_k/\alpha))$ by $1/(\pi^2 \delta_k)$; the error is $O(|\delta_k|)$. The mean-value integral on the short real segment yields, uniformly for $|y| \leq Y$,

$$|F_{b+iy}(q_k - \delta_k/\alpha) - F_{b+iy}(q_k)| \leq C_Y |\delta_k| q_k^D e^{-bq_k}.$$

After division by δ_k , this has a summable majorant. Replacing $|\delta_k|^{-1}$ by q_{k+1} adds at most the bound in (4.5). This proves (4.3) and (4.4). $\square$

4.2 The classification theorem

Theorem 4.3 (All ordinary critical-line convergence types). *At every $z = b + iy$, the following statements hold.*

- (i) *The increasing-action series $P_T(z)$ converges if and only if $A_k \rightarrow 0$. Its value, when it converges, is $\mathcal{S}(z)$.*
- (ii) *It converges absolutely, equivalently unconditionally as a complex series, if and only if $\sum_k A_k < \infty$.*
- (iii) *The two separately indexed lattice series in (2.5)–(2.6) both converge if and only if*

$$\sum_k \eta_k A_k e^{-iyq_k} \quad \text{converges.} \quad (4.6)$$

Either separate lattice series converges precisely under this same criterion. When they converge, their sum equals $\mathcal{S}(z)$.

If $A_k \rightarrow 0$, increasing-action convergence is uniform on each compact segment of $\operatorname{Re} z = b$, and on each compact rectangle $b \leq \operatorname{Re} z \leq B$, $|\operatorname{Im} z| \leq Y$.

Proof. Use the two-lattice blocks of Section 2; each has at most two nodes and its nodes are consecutive in action order. At a cutoff T , all retained blocks are complete except possibly the last one. The sum of the complete blocks tends locally uniformly to $\mathcal{S}$. The incomplete block, if present, contributes exactly one raw residue. Therefore $P_T(z)$ converges to $\mathcal{S}(z)$ whenever all individual residues tend to zero. Conversely, convergence of an ordinary series forces its individual terms to tend to zero.

Lemma 3.3 identifies the first-lattice term test with $A_k \rightarrow 0$. For the second lattice, the same conclusion follows from Lemma 4.1 and Theorem 4.2. This proves (i). All term moduli on the critical line are independent of y ; the same bounds decrease on moving to the right. The block sum is locally uniform on $\mathbb{H}$. These observations prove the stated uniformity.

The absolute convergence equivalence follows either from Lemma 3.3 and the companion estimates, or directly from extraction: an ℓ^1 perturbation cannot change whether the positive sum of the leading moduli A_k/π^2 is finite. This proves (ii). Finally the deleted remainders and u_k, v_k are absolutely summable, whereas the retained leading subseries are opposite scalar multiples of (4.6). This proves (iii). If the separate sums converge, their increasing-action interleaving has their sum as its limit, which (i) identifies with $\mathcal{S}$. $\square$

Remark 4.4 (What the separate criterion does and does not say). Equation (4.6) is a necessary-and-sufficient criterion in explicit continued-fraction data, not a sufficient Diophantine bound. It reduces the full lattice sums to one sparse signed exponential series, with an absolutely summable remainder. For general y it retains a genuine phase-convergence question. The increasing-action criterion has no such residual phase dependence. A detailed classification of the set of imaginary heights satisfying (4.6) is a further problem, not hidden inside an assertion of absolute convergence.

4.3 The zero and infinite endpoints

The positive finite case is the only nontrivial ordinary boundary case. If $\beta(\alpha) = 0$ and $\operatorname{Re} z = 0$, then

$$|r_n(z)| = \frac{n^D}{\pi |\sin(\pi n \alpha)|} \geq \frac{n^D}{\pi}.$$

Thus both increasing-action and separate ordinary interpretations fail the term test on that entire boundary line. If $\beta(\alpha) = \infty$, there is no finite critical boundary, and the raw terms fail to vanish at every fixed $\operatorname{Re} z > 0$. The open-half-plane statements also follow from (3.3): the exponential root test gives absolute convergence for $\operatorname{Re} z > \beta(\alpha)$ and nonvanishing terms below it. Together these facts and Theorem 4.3 complete the ordinary convergence classification requested in the predecessor's Question 1. No assertion about analytic continuation of $\mathcal{S}$ across $\operatorname{Re} z = 0$ is needed.

5 Prescribing the boundary behavior

5.1 A continued-fraction amplitude construction

Theorem 5.1 (Prescribed critical amplitudes). *Fix $b > 0$ and an integer $D \geq 2$. Start with a finite continued-fraction prefix. At stage k , choose a positive target t_k , and choose the next partial quotient within a bounded distance of*

$$X_k = \frac{t_k e^{bq_k}}{q_k^{D+1}}, \quad (5.1)$$

subject to positivity. One may require a specified parity, since the distance to a positive integer of that parity is bounded once X_k is large. Suppose the chosen target prescription satisfies

$$|\log t_k| = o(q_k) \quad (5.2)$$

along the resulting construction. Then the resulting infinite continued fraction is irrational, has $\beta(\alpha) = b$, and satisfies

$$A_k = t_k + O(q_k^{D+1} e^{-bq_k}), \quad A_k/t_k \longrightarrow 1, \quad \sum_k |A_k - t_k| < \infty. \quad (5.3)$$

The limits are unaffected by any finite initial choices.

Proof. The recurrence and bounded rounding error give

$$q_{k+1} = \frac{t_k e^{bq_k}}{q_k^D} + O(q_k),$$

because $q_{k-1} \leq q_k$ eventually. Multiplying by $q_k^D e^{-bq_k}$ gives the first part of (5.3). Condition (5.2) ensures $X_k \rightarrow \infty$, so the rounding rule is feasible eventually and its error is negligible relative to t_k . Geometric denominator growth makes the error absolutely summable. Taking logarithms yields

$$\frac{\log q_{k+1}}{q_k} = b + \frac{\log t_k - D \log q_k + o(1)}{q_k} \rightarrow b.$$

Corollary 3.2 proves the exponent assertion. An infinite simple continued fraction with positive partial quotients is irrational: its consecutive reduced convergents are distinct, with errors tending to zero and alternating signs. Equivalently, the complete-quotient formulas rule out a rational limit with a fixed denominator. $\square$

The prescription can be an explicit function of the already known pair (k, q_k) ; hence the construction is inductive, not an equation for an unknown irrational. For example $t_k = k^{-r}$ and $t_k = q_k^{-r}/k^s$ satisfy (5.2). For the concrete examples, choose $a_0 = 1$, $a_1 \geq 2$, so $1 < \alpha < 2$.

Corollary 5.2 (Every ordinary boundary type at every positive abscissa). *At each prescribed $b > 0$ there are irrational period ratios exhibiting all three cases in Theorem 4.3. Targets k^{-2} , k^{-1} , and 1 give, respectively, absolute convergence, conditional increasing-action convergence, and failure of the term test. The target k produces an unbounded subsequence of individual residue moduli.*

Proof. The first two claims follow from (5.3); the summable rounding errors cannot change the relevant positive-series tests. For the other two, $A_k \sim t_k$ and the extraction theorem give the asserted nonvanishing or unbounded primitive residues. $\square$

5.2 An increasing-action sum with divergent separate sums

Example 5.3 (Opposite divergent lattice sums). Set $a_0 = 1$, choose all a_j for $j \geq 1$ even, and use the even rounding version of Theorem 5.1 with $t_k = 1/k$ in the tail. Modulo two, the convergent vectors alternate as

$$(p_k, q_k) \equiv \begin{cases} (1, 1), & k \text{ even}, \\ (1, 0), & k \text{ odd}. \end{cases}$$

This follows directly from $(p_{k+1}, q_{k+1}) \equiv (p_{k-1}, q_{k-1}) \pmod{2}$. Consequently $\eta_k = 1$ for every k . At $z = b$, the first separate lattice series equals a positive harmonic-type divergent series plus a convergent remainder and tends to $+\infty$; the second tends to $-\infty$. Nevertheless, $A_k \rightarrow 0$ and $\sum A_k = \infty$, so their increasing-action interleaving converges conditionally to $\mathcal{S}(b)$.

This example is not an assignment of a finite value to $+\infty - \infty$. The increasing-action series is a separately defined ordinary series with its own partial sums. The ordering makes its close-pair cancellations occur before the next large-scale contribution, and the remaining unfinished term tends to zero.

Example 5.4 (Both separate lattice sums conditionally converge). Choose $a_0 = 1$, a_1 even, a_2 odd, and all later partial quotients even. For $k \geq 1$, the parity vectors alternate between $(1, 0)$ and $(0, 1)$. Therefore $\eta_k = (-1)^{k+1}$. Choose the tail quotients by Theorem 5.1 with $t_k = 1/k$, respecting even parity. Then

$$\sum_k \eta_k A_k = \sum_k \frac{(-1)^{k+1}}{k} + \text{an absolutely convergent series}$$

converges. Both separate lattice series converge at b , neither absolutely. Their increasing-action interleaving also converges. The argument uses a summable perturbation of the alternating harmonic series, not an unproved monotonicity assertion about A_k .

5.3 A two-point cluster set for a divergent raw expansion

Proposition 5.5 (Bounded spikes of prescribed height). *Fix $L > 0$. Use the parity pattern of Example 5.4 and the constant target $t_k = L$. At the real critical point $z = b$, the increasing-action partial sums have exactly two finite limit points:*

$$\mathcal{S}(b) \quad \text{and} \quad \mathcal{S}(b) + \frac{L}{\pi^2}. \quad (5.4)$$

They are bounded but do not converge. With the same parity pattern and $t_k = k$, a subsequence of partial sums tends to $+\infty$, while block-complete partial sums still tend to $\mathcal{S}(b)$.

Proof. For large k , q_k and λ'_k form one two-node block, since $|\delta_k| \rightarrow 0$. If k is even, $\delta_k > 0$ and the companion $\lambda'_k < q_k$ is first. If k is odd, the integer action q_k is first. In either case the first residue has sign $(-1)^{p_k+q_k+1} = +1$ in its leading term and equals $A_k/\pi^2 + o(1)$. The second residue cancels that leading term.

Every nonprimitive residue belongs to an absolutely summable family by Lemma 4.1, and hence tends to zero individually. Every primitive residue belongs to its indicated pair, apart from finitely many indices. A raw cutoff differs from its last completed block sum by either zero, a vanishing nonprimitive term, or the first term of a primitive pair. Completed block sums tend to $\mathcal{S}(b)$. For $t_k = L$, the only possible limit points are the two in (5.4), and cuts after and inside the primitive pairs realize both. For $t_k = k$, the same cuts inside the pairs diverge to $+\infty$. No assertion is being made here about the cluster sets at nonzero imaginary heights. $\square$

6 The derivative hierarchy of a raw transseries

A normally convergent block sum is holomorphic, but this does not license differentiating a different, conditionally convergent raw representation term by term at a boundary point.

Theorem 6.1 (Exact threshold for each formal derivative). *Under the hypotheses of Section 4, fix an integer $j \geq 0$. The increasing-action series formed by differentiating every raw term j times converges at $b + iy$ if and only if*

$$q_k^j A_k \longrightarrow 0. \quad (6.1)$$

It converges absolutely if and only if $\sum_k q_k^j A_k < \infty$. Whenever it converges, its value is $\mathcal{S}^{(j)}(b + iy)$, uniformly on compact segments of the critical line. The separate derivative series converge precisely when

$$\sum_k \eta_k q_k^j A_k e^{-iyq_k} \quad \text{converges.} \quad (6.2)$$

Proof. Differentiating $c_\lambda e^{-\lambda z}$ multiplies it by $(-\lambda)^j$. Apart from the common factor $(-1)^j$, the result is exactly the degree- $D + j$ residue family. Its critical amplitude is $q_k^j A_k$. Apply the extraction and classification theorems with $D + j$ in place of D . Proposition 2.2 identifies the corresponding block sum with the derivative of $\mathcal{S}$. $\square$

Corollary 6.2 (Arbitrarily prescribed finite differentiation depth). *Fix an integer $r \geq 0$. There are period ratios with any prescribed $b > 0$ for which derivatives of orders below r have absolutely convergent raw series, order r has a conditionally convergent increasing-action raw series, and order $r + 1$ fails the term test. There are also examples with absolute raw convergence at every derivative order on $\text{Re } z = b$, while the absolute convergence abscissa remains exactly b .*

Proof. For the first statement prescribe $t_k = q_k^{-r}/k$. Then $q_k^j A_k$ differs from q_k^{j-r}/k by a summable, exponentially small sequence for every fixed j . For $j < r$, geometric growth of q_k gives summability. At $j = r$ the amplitudes tend to zero but have divergent sum. At $j = r + 1$, $q_k/k \rightarrow \infty$.

For the second statement prescribe $t_k = e^{-\sqrt{q_k}}$. Condition (5.2) holds, so $\beta(\alpha) = b$. For each fixed j , $\sum q_k^j e^{-\sqrt{q_k}} < \infty$. Theorem 6.1 gives the assertion. $\square$

The failure in the first statement belongs to the representation, not to the analytic function: $\mathcal{S}$ has derivatives of all orders near every point with positive real part. At a high enough order the raw termwise derivative ceases to be an ordinary convergent series, but the block derivative remains valid. This separates analytic regularity from raw-sector regularity without changing the formal action set.

7 Riesz cutoff regularization without Diophantine assumptions

We now return to d lattices, the period box (2.7), and $D \geq d$. No approximation exponent is assumed finite. Positive poles are simple in the principal theorem; rational confluence is treated separately in Section 8.

7.1 A weighted cutoff that can split a block

For an integer $m \geq 1$, define the finite Riesz mean

$$\mathcal{R}_T^{(m)}(z) = \sum_{\lambda < T} \left(1 - \frac{\lambda}{T}\right)^m c_\lambda e^{-\lambda z}, \quad T > 0. \quad (7.1)$$

This is the usual Riesz weight for a Dirichlet series indexed by the frequency λ ; see [5] for general context. Our proof is a direct local argument for the present meromorphic kernel. In particular, it does not assume that the raw series has a nonempty half-plane of convergence.

Let $w_{T,m}(t) = T^{-m}(T-t)_+^m$ on the real line, where $(u)_+ = \max(u, 0)$. For a simple block $C = \{\lambda_1, \dots, \lambda_r\}$, its contribution to (7.1) is exactly

$$[\lambda_1, \dots, \lambda_r](g_C(\cdot, z)w_{T,m}). \quad (7.2)$$

The identity remains valid when the cutoff lies between nodes of that block. It is the central reason the pole gaps need not enter the bound.

Lemma 7.1 (A cutoff through a cluster). *Suppose a block of $r \leq d$ distinct nodes straddles $T \geq 1$ and $m \geq r - 1$. For $K \in \mathbb{H}$,*

$$\sup_{z \in K} |[\lambda_1, \dots, \lambda_r](g_C(\cdot, z)w_{T,m})| \leq C_K T^{D-m} e^{-\sigma T}, \quad \sigma = \min_K \operatorname{Re} z. \quad (7.3)$$

The same estimate holds with the ordinary polynomial $T^{-m}(T-t)^m$ in place of $w_{T,m}$ on that block.

Proof. The block interval has bounded length $W = (d-1)h$, so $|t - T| \leq W$ there. For $0 \leq j \leq r-1 \leq m$, the weak derivatives satisfy

$$|w_{T,m}^{(j)}(t)| \leq (m)_j T^{-m} W^{m-j}$$

almost everywhere, with harmless replacement by a larger constant if $W < 1$. Here $(m)_j = m!/(m-j)!$. In particular, $w_{T,m} \in W^{r-1,\infty}$, including the endpoint case $m = r-1$. Leibniz's rule and (2.9) bound the $(r-1)$ st weak derivative of the product by $C_K T^{-m}(1+T)^D e^{-\sigma T}$. The Sobolev divided-difference bound (2.11) proves (7.3). For the ordinary polynomial the same argument is smooth and identical. $\square$

7.2 The finite derivative bias and the exponential remainder

Theorem 7.2 (Uniform Riesz recovery and its exact bias). *Let $m \geq d - 1$ be an integer. For every compact $K \Subset \mathbb{H}$, there is a constant depending only on K, d, D, m, m_0, M_0 such that, for all $T \geq 1$ and every period vector in the box with simple positive poles,*

$$\sup_{z \in K} \left| \mathcal{R}_T^{(m)}(z) - \sum_{j=0}^m \binom{m}{j} \frac{\mathcal{S}^{(j)}(z)}{T^j} \right| \leq C_K T^{D-m} e^{-\sigma T}, \quad \sigma = \min_K \operatorname{Re} z. \quad (7.4)$$

Consequently $\mathcal{R}_T^{(m)} \rightarrow \mathcal{S}$ locally uniformly on $\mathbb{H}$. The bound does not depend on any internal pole separation or on an exponential approximation exponent.

Proof. First weight every complete block by the ordinary polynomial $(1 - t/T)^m$, even for blocks above the cutoff. Its normally convergent sum is

$$\begin{aligned} Q_T(z) &= \sum_C [\lambda_1, \dots, \lambda_r] (g_C(\cdot, z) (1 - t/T)^m) \\ &= \sum_{j=0}^m \binom{m}{j} \frac{\mathcal{S}^{(j)}(z)}{T^j}. \end{aligned} \quad (7.5)$$

Indeed the coefficient $(-t)^j$ in the polynomial is precisely the factor produced by differentiating e^{-zt} j times. The derivative normal convergence in Proposition 2.2 justifies the finite expansion and the block summation.

For blocks wholly below T , the polynomial weight equals the truncated weight. There is at most one block straddling T . Both of its weighted contributions are bounded by Lemma 7.1, since $m \geq d - 1 \geq r - 1$.

It remains to bound polynomial-weighted blocks wholly above T . For such a block with minimum $a \geq T$, Leibniz's rule, the fixed interval length, and (2.9) give

$$|[\lambda_1, \dots, \lambda_r] (g_C(\cdot, z) (1 - t/T)^m)| \leq C_K T^{-m} (1 + a)^D (1 + a - T)^m e^{-\sigma a}. \quad (7.6)$$

Bin the minima in $[T + \ell, T + \ell + 1)$, using their bounded count. Because $T \geq 1$, the sum of (7.6) is bounded by

$$C_K T^{D-m} e^{-\sigma T} \sum_{\ell \geq 0} (1 + \ell)^{D+m} e^{-\sigma \ell}.$$

The last series is finite. Absorb it in the constant. Combining this tail with the possible split block proves (7.4). The derivative terms of positive order have factors T^{-j} and are uniformly bounded on K , which gives convergence to $\mathcal{S}$. $\square$

Remark 7.3 (What uniformity means). The period box may contain rational coincidences, while the principal formula (7.1) is initially evaluated on its simple-pole locus. The estimate is uniform as those coincidences are approached. It asserts a bound, not continuity of every finite cutoff at a coincidence situated exactly at T . That distinction is essential at minimal order. Also, the power T^{D-m} may be negative when $m > D$; there is no restriction $m \leq D$.

7.3 The two-lattice Cesàro interpretation

For simple poles, a finite interchange gives the exact identity

$$\mathcal{R}_T^{(1)}(z) = \frac{1}{T} \int_0^T P_u(z) du. \quad (7.7)$$

Thus for two lattices the continuous first Cesàro mean of the raw cutoff converges everywhere in $\mathbb{H}$, even when the raw cutoff fails the term test there. A huge unmatched residue persists

only over the very small interval between its two actions. The factor $(1 - \lambda/T)$ integrates that interval rather than estimating its height alone. Equation (7.4) is stronger than mere Cesàro convergence: it gives the exact T^{-1} bias and an exponentially small remaining error.

This interpretation does not say that a naive floating-point evaluation of (7.1) is well conditioned. Tiny sine denominators and cancellation can still require high precision or stable divided differences. The theorem controls the mathematical finite sum, not the roundoff error of an arbitrary implementation.

8 Sharp cutoff order, confluence, and exponential acceleration

8.1 The gap-independent threshold is sharp

Theorem 8.1 (Minimal order for uniform collision control). *For $d \geq 2$, an integer order $m < d - 1$ cannot give a bound uniform across all simple period vectors in a fixed compact positive box, even at a fixed positive cutoff and a fixed $z \in \mathbb{H}$. More precisely, take*

$$\lambda_1 = 1 - \varepsilon, \quad \lambda_j = 1 + (j - 1)\varepsilon \quad (2 \leq j \leq d), \quad \omega_j = \lambda_j^{-1}, \quad T = 1, \quad (8.1)$$

with positive irrational $\varepsilon \rightarrow 0$. Then the positive pole sets are pairwise disjoint and

$$\mathcal{R}_1^{(m)}(z) = -\frac{e^{-z}}{\pi^d \prod_{j=2}^d j} \varepsilon^{m-d+1} (1 + O_z(\varepsilon)). \quad (8.2)$$

Thus $d - 1$ is the minimal integer order for the gap-independent uniform control proved in Theorem 7.2.

Proof. For small ε , only the first pole λ_1 is below $T = 1$; even its second lattice multiple is above one. Distinct ratios $(1 + c_i\varepsilon)/(1 + c_j\varepsilon)$ with different rational c_i, c_j cannot be rational when ε is irrational, so the positive pole sets are disjoint.

At λ_1 , the derivative of its own sine factor is $-\pi\omega_1$, while for $j \geq 2$,

$$\sin(\pi\omega_j\lambda_1) = \pi j\varepsilon(1 + O(\varepsilon)).$$

Also $\lambda_1^D = 1 + O(\varepsilon)$ and $e^{-z\lambda_1} = e^{-z}(1 + O_z(\varepsilon))$. The residue formula, multiplied by $(1 - \lambda_1)^m = \varepsilon^m$, gives (8.2). Its coefficient is nonzero, proving the claimed failure for $m < d - 1$. $\square$

The sharpness just proved is *parameter-uniform*. It must not be misread as a proof that every fixed d -lattice input needs order $d - 1$. Some inputs already have absolutely convergent raw expansions. For $d = 2$, examples with $\beta = \infty$ do show that order zero can fail everywhere in $\mathbb{H}$ for one fixed irrational input. A sharp fixed-input classification for more lattices is a separate question.

8.2 Boundedness is not cutoff continuity

Proposition 8.2 (One additional order for full cutoff confluence). *At a block of multiplicity at most d , the finite-cutoff expressions have gap-independent bounds when $m \geq d - 1$. If $m \geq d$, they also extend continuously as the nodes coalesce, including when their common limit equals T . At $m = d - 1$, continuity can fail at that cutoff.*

Proof. The boundedness is Lemma 7.1. If $m \geq d$, then $t \mapsto (T - t)_+^m$ is C^{d-1} . Divided differences of order at most $d - 1$ of the smooth regular factor times this weight have their continuous confluent limits. This follows from the simplex formula and continuity of the required derivative, even when the limiting node equals the cutoff.

For $m = d - 1$, consider a d -node cluster approaching T with a limiting regular factor $g(T, z) \neq 0$. If all nodes approach from above, its truncated weighted contribution is zero. If all approach from below, its limit is

$$\frac{1}{(d-1)!} \partial_t^{d-1} (g(t, z)(1 - t/T)^{d-1}) \Big|_{t=T^-} = \frac{(-1)^{d-1}}{T^{d-1}} g(T, z),$$

which is nonzero. Such configurations are obtained by taking the first poles of the sine lattices close to T . Intermediate configurations can give other limits, as the divided difference of a truncated power depends on how the cutoff cuts the collapsing nodes. $\square$

At a coincident pole away from the cutoff, the correct definition is

$$\text{Res}_{t=\lambda} \left((1 - t/T)^m e^{-zt} \hat{H}(t) \right), \quad \lambda < T, \quad (8.3)$$

not $(1 - \lambda/T)^m$ times the residue of $e^{-zt} \hat{H}(t)$. The two formulas agree at a simple pole but need not agree at a multiple pole, because derivatives of the weight contribute. For $m \geq d$, excluding a pole exactly at T is compatible with the continuous extension, since all derivatives through order $d - 1$ of the cutoff weight vanish there. At $m = d - 1$ no convention at that exact pole can make every approach continuous. Uniform exponential error control survives this local ambiguity, but finite-cutoff continuity does not.

8.3 Removing the bias with finitely many cutoffs

Corollary 8.3 (Exponentially accurate extrapolated action cuts). *For $m \geq d - 1$, define*

$$w_\ell = \frac{(-1)^{m+1-\ell} \ell^m}{(\ell-1)!(m+1-\ell)!}, \quad 1 \leq \ell \leq m+1, \quad (8.4)$$

$$\mathcal{A}_T^{(m)}(z) = \sum_{\ell=1}^{m+1} w_\ell \mathcal{R}_{\ell T}^{(m)}(z). \quad (8.5)$$

Then, with the same uniformity as in Theorem 7.2,

$$\sup_{z \in K} |\mathcal{A}_T^{(m)}(z) - \mathcal{S}(z)| \leq C_K T^{D-m} e^{-\sigma T}. \quad (8.6)$$

In particular,

$$\mathcal{A}_T^{(1)} = 2\mathcal{R}_{2T}^{(1)} - \mathcal{R}_T^{(1)}, \quad (8.7)$$

$$\mathcal{A}_T^{(2)} = \frac{1}{2}\mathcal{R}_T^{(2)} - 4\mathcal{R}_{2T}^{(2)} + \frac{9}{2}\mathcal{R}_{3T}^{(2)}. \quad (8.8)$$

Proof. The weights are the Lagrange interpolation weights for evaluating a polynomial of degree at most m at zero from its values at $1, 1/2, \dots, 1/(m+1)$. Indeed the weight at $1/\ell$ is

$$\prod_{r \neq \ell} \frac{-1/r}{1/\ell - 1/r} = \frac{(-1)^{m+1-\ell} \ell^m}{(\ell-1)!(m+1-\ell)!}.$$

Therefore $\sum w_\ell = 1$ and $\sum w_\ell \ell^{-j} = 0$ for $1 \leq j \leq m$. Apply these identities to the derivative polynomial in (7.4). Its nonconstant terms cancel exactly. Each remaining error is bounded by $C_K(\ell T)^{D-m} e^{-\sigma \ell T}$, and the finite weighted sum has the bound in (8.6). $\square$

The acceleration is algebraic and exact, not a fitted numerical extrapolation. It uses $m + 1$ cutoffs and no derivative values. Its largest action is $(m + 1)T$, so enumeration is linear in T for fixed d, m before arithmetic precision is considered. An effective bit-cost claim would additionally require certified representations of the periods, reliable pole comparisons, and stable evaluation of the retained finite sums. Those requirements are not supplied merely by an analytic convergence theorem.

9 Transport through a nonlinear inverse

The predecessor already provides an analytic inverse calculus after block summation [1]. The result here is a quantitative compatibility statement for the new raw-data regularization: one may replace the block sum by an extrapolated finite Riesz sum and then invert, without losing exponential accuracy.

Theorem 9.1 (Certified local inverse stability). *Fix a closed disk $\overline{D} = \{z : |z - z_0| \leq \rho\}$ and a larger concentric disk $\overline{D}^+$ of radius $\rho + \eta$, where $\eta > 0$ and $\overline{D}^+ \subset \mathbb{H}$. Put*

$$M = \sup_{\overline{D}} |\mathcal{S}|, \quad L = \sup_{\overline{D}} |\mathcal{S}'|, \quad |\varepsilon|M \leq \rho/4, \quad |\varepsilon|L \leq 1/4. \quad (9.1)$$

For each $|w - z_0| \leq \rho/2$, the equation

$$z + \varepsilon \mathcal{S}(z) = w \quad (9.2)$$

has a unique solution in $\overline{D}$. For all sufficiently large T , the equation

$$z_T + \varepsilon \mathcal{A}_T^{(m)}(z_T) = w \quad (9.3)$$

also has a unique solution there, where $m \geq d - 1$. If $\delta_T = \sup_{\overline{D}^+} |\mathcal{A}_T^{(m)} - \mathcal{S}|$, then

$$|z_T - z| \leq \frac{|\varepsilon|}{1 - |\varepsilon|L} \delta_T \leq \frac{4|\varepsilon|}{3} \delta_T. \quad (9.4)$$

In particular, the inverse error is $O(|\varepsilon|T^{D-m}e^{-\sigma T})$ uniformly in these targets, where $\sigma = \min_{\overline{D}^+} \operatorname{Re} z$.

Proof. The map $z \mapsto w - \varepsilon \mathcal{S}(z)$ sends $\overline{D}$ into its concentric disk of radius $3\rho/4$. On the convex disk its Lipschitz constant is at most $|\varepsilon|L \leq 1/4$. The contraction theorem gives the unique exact solution.

Corollary 8.3 makes $\delta_T \rightarrow 0$. Cauchy's estimate on circles of radius η centered in $\overline{D}$ gives

$$\sup_{\overline{D}} |(\mathcal{A}_T^{(m)})' - \mathcal{S}'| \leq \delta_T/\eta.$$

For large T , require $|\varepsilon|\delta_T \leq \rho/8$ and $|\varepsilon|\delta_T/\eta \leq 1/4$. Then $z \mapsto w - \varepsilon \mathcal{A}_T^{(m)}(z)$ maps $\overline{D}$ into its disk of radius $7\rho/8$ and has Lipschitz constant at most $1/2$. It too has a unique fixed point.

Subtract the two fixed-point equations and insert $\mathcal{S}(z_T)$:

$$|z_T - z| \leq |\varepsilon|L|z_T - z| + |\varepsilon|\delta_T.$$

Solving this scalar inequality proves (9.4), and (8.6) gives the final rate. The argument also shows, by the holomorphic implicit function theorem, that the local inverse branches are holomorphic in the interior target disk. $\square$

The factor $2\pi i$ converting $\mathcal{S}$ into a Stokes jump can be absorbed in ε . More general exactly invertible cores can be treated in core coordinates, provided the image disks and derivative bounds are supplied. That coordinate reduction and the abstract contraction estimate are standard; the substantive new input is the uniform exponentially accurate approximation in Corollary 8.3.

The theorem concerns a local analytic inverse. It does not assert global univalence, a complete Borel singularity description for the inverse, or convergence of its fully expanded raw action series. In particular, when $\beta = \infty$, the unregularized linear residue contribution may fail the term test, while the regularized finite inverse converges to the correct local branch.

10 Reproducible checks and numerical behavior

10.1 What was actually checked

The supplied `verify.py` was executed with Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0 at 110 decimal digits. It passed **192 exact assertions** and **49 numerical assertions**. The exact checks are distributed as follows:

Finite identity family	Assertions
Richardson moments, orders 1 through 8	44
Continued-fraction determinant and complete-quotient identities	36
All-even parity pattern	31
Alternating parity pattern	30
Truncated-power collision scaling, $d = 2, \dots, 7$	18
Divided-difference polynomial reproduction	27
Weighted derivative identities	6
Total	192

The numerical checks comprise two reference-cutoff comparisons, ten finite-bias remainders, ten extrapolated remainders, four cutoff continuity diagnostics, fifteen actual sine-kernel collision scalings, and eight finite Legendre classifications encountered up to $n = 500$. All tolerances and all measured values are recorded in `results/verification.json`.

These are finite algebra checks and high-precision diagnostics, not directed-rounding interval certificates. In particular, the program does not approximate an entire rapidly growing continued-fraction construction and infer its infinite exponent from a finite prefix. The exponent and the endpoint types of those constructions are proved in Section 5. The program also does not turn the analytic tail constants or the nonlinear inverse theorem into machine-checked proofs.

10.2 Derivative-bias removal

Table 1 uses $z = 0.8$. For $d = 2$, the periods are $(1, \sqrt{2})$, $D = 2$, and $m = 1$. For $d = 3$, they are $(1, \sqrt{2}, \sqrt{3})$, $D = 3$, and $m = 2$. Define

$$Q_T(z) = \sum_{j=0}^m \binom{m}{j} \frac{\mathcal{S}^{(j)}(z)}{T^j}.$$

The reference sums and derivatives use poles below 260; comparisons with a cutoff at 220 differed by less than 10^{-63} for the sum in both models. This is a finite reference comparison, not a proof of a 10^{-63} tail certificate. The displayed discrepancies are many orders of magnitude larger than that comparison scale.

The uncorrected error retains the algebraic derivative bias. Removing that bias exposes the exponential scale predicted by Theorem 7.2. There is no theorem that the error must decrease monotonically at every small cutoff: a moving cutoff can cross or split a block, and finite terms can cancel. The theorem is the uniform upper estimate, not a pointwise monotonicity rule.

10.3 A collision exactly at the cutoff

Set $D = 2$, $z = 1.7$, $T = 2$, and

$$\alpha_{\pm}(k) = \frac{3}{2} \pm \sqrt{2} 10^{-k}.$$

The actions 2 and $3/\alpha$ collide at the cutoff as $k \rightarrow \infty$. Let $\mathcal{R}_{2,\pm}^{(m)}$ denote the two finite sums. For $m = 0$, use the raw cutoff P_2 . Table 2 illustrates all three phenomena in Section 8: the raw

d	m	T	$ \mathcal{R}_T^{(m)} - \mathcal{S} $	$ \mathcal{R}_T^{(m)} - Q_T $	$ \mathcal{A}_T^{(m)} - \mathcal{S} $
2	1	8	5.1888×10^{-4}	3.9259×10^{-4}	3.9543×10^{-4}
2	1	12	5.0987×10^{-5}	3.3205×10^{-5}	3.3214×10^{-5}
2	1	16	6.1721×10^{-5}	1.4221×10^{-6}	1.4221×10^{-6}
2	1	24	4.21×10^{-5}	4.4517×10^{-9}	4.4517×10^{-9}
2	1	32	3.1572×10^{-5}	6.9309×10^{-12}	6.9309×10^{-12}
3	2	8	4.538×10^{-6}	2.0923×10^{-5}	1.0766×10^{-5}
3	2	12	6.3028×10^{-6}	4.3681×10^{-6}	2.1823×10^{-6}
3	2	16	7.9841×10^{-6}	7.5645×10^{-8}	3.7821×10^{-8}
3	2	24	5.2087×10^{-6}	4.3439×10^{-10}	2.172×10^{-10}
3	2	32	3.8831×10^{-6}	5.2882×10^{-13}	2.6441×10^{-13}

Table 1: Finite Riesz bias and its extrapolated removal at $z = 0.8$. Ordinary high-precision arithmetic; no interval enclosure is implied.

k	$P_{2,+}$	$\mathcal{R}_{2,-}^{(1)}$	$\mathcal{R}_{2,+}^{(1)}$	$ \mathcal{R}_{2,+}^{(2)} - \mathcal{R}_{2,-}^{(2)} $
4	47.805	-9.352226×10^{-3}	-4.842602×10^{-3}	1.1465×10^{-6}
8	4.782×10^5	-9.351778×10^{-3}	-4.843219×10^{-3}	1.1465×10^{-10}
16	4.782×10^{13}	-9.351778×10^{-3}	-4.843219×10^{-3}	1.1465×10^{-18}
32	4.782×10^{29}	-9.351778×10^{-3}	-4.843219×10^{-3}	1.1465×10^{-34}

Table 2: A cutoff collision distinguishes boundedness from continuity.

cutoff becomes large; the first-order cutoff remains bounded but has different limits from the two sides; the second-order difference tends to zero.

At $\alpha = 3/2$ itself, the sine kernel has a double pole at action two. Using a simple-pole formula there would be invalid. The program avoids that error by using irrational perturbations for the table. The analytic discussion explains the correct multiple-pole residue and why order one cannot have a side-independent cutoff limit.

10.4 Rebuilding and the computational boundary

The package includes the recorded table inputs, so the article can be compiled without rerunning the computation. The commands

```
python verify.py --outdir build/results
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

regenerate checks in a separate directory and typeset the recorded article. The supplied build script directs TeX auxiliary output to `build/`. No network calls or external data are used by the verification program. Its simple-residue evaluator rejects a sine phase resolved as an integer, including the tested exact coincidences; it is a diagnostic implementation, not a general stable confluent-residue library.

A production numerical method should use the divided-difference or contour representation for nearly colliding nodes, evaluate tail bounds with outward rounding, and certify comparisons against the cutoff. The present numerical tables should not be represented as having performed those additional tasks.

11 Further research questions and proposed projects

The first question in the predecessor’s list—ordinary convergence on the critical line—has been answered above. The following projects start at the remaining boundaries of the proved results rather than restating that question unchanged.

1. The phase spectrum of the separate lattice sums

For an explicitly prescribed amplitude sequence and parity pattern, determine the set

$$E = \left\{ y \in \mathbb{R} : \sum_k \eta_k A_k e^{-iyq_k} \text{ converges} \right\}.$$

Theorem 4.3 gives an exact reduction to this set, not its measure, category, or Hausdorff dimension. A useful next result would compare almost-everywhere and exceptional-point convergence for superlacunary denominators while retaining the continued-fraction realizability constraint.

2. Fixed-input minimal smoothing for three or more lattices

Theorem 8.1 proves the sharp order for estimates uniform in the periods. For a fixed vector, smaller order may suffice. Seek a necessary-and-sufficient criterion expressed through simultaneous near-collisions and the effective multiplicities they create at large action. This requires more than applying a two-lattice exponent pairwise, because a cutoff can split a cluster of several poles.

3. Fractional cutoff orders and the optimal boundary profile

The local scaling obstruction extends naturally to noninteger cutoff orders, but the finite polynomial bias is special to integer m . Develop a version for $(1 - \lambda/T)_+^s$ with real s , giving its sharp regularity threshold and a uniform bias expansion. Determine which fractional combinations can improve error or reduce cancellation without sacrificing confluence control.

4. Countably many lattices and unbounded cluster size

For infinitely many period lattices the pigeonhole argument no longer bounds cluster cardinality. Find a weighted criterion balancing the number of nodes, regular-factor norms, and action growth. The aim is a genuine countable-lattice theorem, not an invocation of Theorem 7.2 outside its fixed- d hypothesis.

5. Algebraic and logarithmic Borel singularities

Replace meromorphic poles by branch singularities and identify a cutoff regularity rule that plays the role of multiplicity minus one. Such a theorem must specify branches and continuation paths, replace residues by Hankel contributions, and control merging singularities. It cannot be obtained merely by substituting a noninteger pole order in the present residue formulas.

6. Compatibility with transseries products and composition

The local inverse theorem uses uniform approximation of one analytic function. A broader algebraic objective is a summation map compatible with products, differentiation, and composition on an explicit class of raw action expansions. Identify the topology in which the smoothed cutoffs define that map and determine when its actionwise coefficients agree with formal operations. Local inverse stability alone does not settle this algebraic compatibility.

7. Certified adaptive action budgets and bit complexity

Make the constants in (2.9) and (8.6) effective and computationally sharp for specified period boxes. Combine interval representations of the periods, certified cluster comparisons, confluent divided differences, and outward-rounded tails. A useful output is an end-to-end algorithm whose bit cost includes near-collision precision, not just the number of retained actions.

8. The line of zero real part after regularization

For the present $D \geq d$, ordinary raw convergence on $\operatorname{Re} z = 0$ already fails the term test. Investigate what boundary values, analytic continuations, or distributional limits the Riesz sums possess there. The exponential majorants used in this article disappear at that line; a new oscillatory or distributional argument is required.

9. General multipliers and logarithmic coefficient growth

A holomorphic multiplier with uniform tube bounds can replace t^D in parts of the block argument. Determine the endpoint extraction and sharp Riesz tail for multipliers such as $t^D(\log t)^r$ or regularly varying functions. Zeros of the multiplier at convergent actions may alter the amplitude test, so a correct theorem must include nonvanishing or explicit cancellation hypotheses.

10. A staged formalization in the repository

The arithmetic and analytic halves can be developed independently. First formalize the weighted denominator test and the primitive-pair extraction. Separately formalize divided differences of truncated powers with bounds that do not mention node separation. Then connect the finite polynomial identity to block sums and package the inverse stability estimate. The proof-dependency map below supplies concrete milestones. No claim is made that these new theorems are already available in Lean or that the required analytic interfaces have been checked in the current Mathlib version.

12 Proof audit, scope, and conclusions

12.1 Dependency map

The main logical dependencies are short enough to record explicitly.

Result	Essential inputs
Critical absolute and term tests	Best approximation, exponential weights, consecutive denominator intervals.
Separate-series extraction	Legendre reduction, finite exponential exponent, exact companion-pole identity, summable errors.
Increasing-action classification	Two-node blocks, the individual term test, the normally convergent block sum.
Prescribed examples	Continued-fraction recurrence, bounded parity rounding, a summable realization error.
Riesz recovery	Bounded cluster size, pole-cancelled tube bounds, Sobolev divided differences, a polynomial weighted tail.
Sharpness and confluence	A first-pole collision family and the exact regularity of the truncated power at its cutoff.
Acceleration and inverse transport	Finite interpolation moments, uniform approximation, Cauchy estimates, contraction.

None of the main existence or convergence claims relies on a numerical plot or on the observed behavior of a long but finite continued fraction. Conversely, the exact finite identities checked by the program do not verify all hypotheses of an infinite analytic theorem. The final article is a conventional proof document, not a formal proof assistant artifact.

12.2 Claims not made

The source comparison covers the named predecessor, its local statements and research question, and the cited inventory. It is not a claim that the entire repository was audited theorem by theorem. The general theory of Riesz summability, the irrationality-base formula, the divided-difference method, and the abstract local inverse estimate are credited rather than relabeled as discoveries.

For fixed finite sine lattices, the new uniform Riesz theorem is unconditional on Diophantine approximation. The endpoint classification, however, concerns exactly two irrational lattices. No extension to arbitrary countable singular supports, no full inverse-resurgence theorem, and no global inverse univalence is asserted. The sharp cutoff-order claim is explicitly about period-uniform gap-independent control. Independent research priority and peer review remain separate from the correctness of the written arguments.

12.3 Conclusion

The critical line is governed by a finer invariant than its exponential abscissa. Continued fractions supply that invariant in the sequence $q_k^D q_{k+1} e^{-bq_k}$. Vanishing of these amplitudes controls unfinished pairs; summability controls absolute convergence; their parity and phases control the two separate lattice sums. This separates three questions that the open-half-plane root test cannot distinguish.

The cutoff theorem provides a complementary answer that does not ask for small-divisor arithmetic at all. A truncated power with enough regularity controls the divided differences of a block even when the cutoff splits it. Its finite derivative bias can be cancelled exactly, yielding an exponentially accurate finite-action representation and a stable nonlinear inverse approximation. In this setting, both the arithmetic boundary and the analytic regularization can therefore be made precise without identifying a formal transseries with an unconditionally convergent raw numerical sum.

A Repository and primary-literature provenance

The pinned repository commit is

3d5973524506411392a911470b5ddc35521568ea.

The directly inspected predecessor source is

`Analysis/Transseries/docs/series-and-transseries/Resonance_Block_Summation
_Transseries/article.tex.`

Its Git blob identifier is 934422729a618bf2bb6792f4341ec73b9f5cf94e. The inspected ranges include lines 1–240 (scope and conventions), 440–710 (divided differences, block summation, residues, abscissa), 1110–1370 (inverse transport and the general bounded-cluster criterion), and 1371–1580 (stable evaluation and the explicit boundary research question). Its package README and the group inventory were also read through the connected GitHub tools. The repository was not modified.

For the external primary sources, Sondow’s continued-fraction formula and Legendre lemma were checked in the PDF, including the displayed formula for the logarithm of the irrationality

base. De Boor’s Genocchi–Hermite formula and the contour/divided-difference discussion were inspected. The Defant–Schoolmann paper was used for the established Riesz-summability context, not as a black-box proof of Theorem 7.2. These references establish antecedents and notation; a comprehensive priority review of all specialized small-divisor and summability literature has not been performed.

References

- [1] *ProveIt* repository, *Resonance-Block Summation of Transseries: Small divisors, coalescing Stokes actions, and nonlinear inversion*, research report filed 29 September 2026, with editorial amendments. Source pinned at commit 3d5973524506411392a911470b5ddc35521568ea. Especially `thm:block`, `thm:arithmetic`, `thm:inverse`, and further-research Question 1. https://github.com/VladimirReshetnikov/ProveIt/blob/3d5973524506411392a911470b5ddc35521568ea/Analysis/Transseries/docs/series-and-transseries/Resonance_Block_Summation_Transseries/article.tex.
- [2] *ProveIt* repository, *Series and transseries*, group inventory, `Analysis/Transseries/docs/series-and-transseries/README.md`, same pinned commit, inspected 29 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/3d5973524506411392a911470b5ddc35521568ea/Analysis/Transseries/docs/series-and-transseries/README.md>.
- [3] J. Sondow, *Irrationality Measures, Irrationality Bases, and a Theorem of Jarník*, arXiv:math/0406300 (2004). See Lemma 1 and Theorem 1, especially equation (3). <https://arxiv.org/abs/math/0406300>.
- [4] C. de Boor, *Divided Differences*, *Surveys in Approximation Theory* **1** (2005), 46–69. See the contour representation and the Genocchi–Hermite formula (52). <https://arxiv.org/abs/math/0502036>.
- [5] A. Defant and I. Schoolmann, *Riesz Summability on Boundary Lines of Holomorphic Functions of Finite Order Generated by Dirichlet Series*, arXiv:2107.10145, version 3 (2022). The quoted title is the title printed in the inspected PDF. <https://arxiv.org/abs/2107.10145>.